Prabhath Vinay Vipparthi

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SUMMARY

MS Data Science graduate (NJIT, GPA 3.7, May 2026) who builds and ships production machine learning and GenAI systems end to end, from data and feature engineering through model training, deployment, monitoring, and explainability. Work spans a deployed financial-intelligence platform with automated ingestion and full model-output monitoring; an LLM instruction-tuning pipeline built on the StarCoder2 methodology over 30,000+ source files; explainable ML on 20,000-record datasets with SHAP attribution; and formally powered statistical experimentation on roughly 96,000 real customers. Strong in Python, SQL, and PostgreSQL across the full model lifecycle. Work authorization: F-1 OPT, STEM-eligible.

EDUCATION

Master of Science (MS) in Data Science, Computational Track — NJIT, Ying Wu College of Computing
May 2026 | Newark, NJ | GPA 3.7

B.Tech Electrical & Electronics Engineering — Jawaharlal Nehru Technological University, Kakinada (JNTUK)
Apr 2023 | Andhra Pradesh, India

EXPERIENCE

Graduate Data Science Capstone— NJIT, Learning & Development Initiative (LDI)
Jan 2026 - May 2026 | Newark, NJ

- Engineered a production AI-assisted digital badge classification system (FastAPI, React/Vite, spaCy, SQLite) automating NJIT's institutional taxonomy across three dimensions (category, type, cognitive level), correctly classifying all 20 held-out real-world submissions against ground-truth labels from 5 institutional units, with confidence scoring on every output.
- Built a deterministic 3-stage rule engine (19 rules) with full rule traceability, 8-element plain-English decision explanations, immutable governance audit logs, and human-in-the-loop override workflows requiring written justification, making every classification explainable and auditable.
- Delivered a 351-test automated validation suite (100% pass rate) enforcing data quality across unit, integration, and 7 end-to-end workflow scenarios covering NLP extraction accuracy, taxonomy classification consistency, and metadata ingestion correctness.

PROJECTS

FinSight — Production Pre-Market Intelligence Platform | *Python, FastAPI, PostgreSQL/pgvector, GitHub Actions, Render*

- Built and deployed a production financial-briefing service that runs automatically every weekday via GitHub Actions cron, ingesting roughly 130 market symbols (futures, macro indicators, sector ETFs, watchlist) with a keyed Polygon.io to yfinance fallback for resilience.
- Instrumented full model-output monitoring, logging every generation with prompt, output, model, latency, and status, with automatic fallback so the briefing always ships; this is the production observability pattern needed for decision systems.
- Added a retrieval-augmented Q&A layer (pgvector embeddings with date-cited answers) and 68 automated tests across 9 modules running offline in CI.

StarCoder2 Self-Alignment Pipeline — LLM Instruction-Tuning Dataset | *Python, TypeScript, Tree-sitter, vLLM, StarCoder2-3B, Hugging Face*

- Implemented the SelfOSSInstruct methodology from the StarCoder2 technical report to build a code-generation instruction-tuning dataset through a 3-stage pipeline: seed extraction with quality filtering, concept and instruction generation, and response generation with model-based quality scoring.
- Processed 30,000+ TypeScript files from The Stack v2 with Tree-sitter AST parsing and the TypeScript compiler API, filtering to 5,791 quality-validated seed functions; ran StarCoder2-3B via vLLM batched inference (batch size 32) on a Colab T4 GPU to generate concepts, instructions, and response pairs.

Financial Risk — Explainable Loan Approval Modeling | *Python, Scikit-learn, TensorFlow, SMOTE, SHAP, GridSearchCV*

- Built a credit-risk classification pipeline on 20,000 loan applications (36 features, 76% class imbalance); applied SMOTE on training folds only to prevent data leakage and compared 6 models (LR, DT, RF, SVM, KNN, ANN) with 5-fold cross-validated GridSearch tuning across F1, AUC-ROC, and precision-recall.
- Applied model-agnostic SHAP across all 6 models to surface CreditScore, AnnualIncome, and DebtToIncomeRatio as top risk drivers, producing compliance-ready feature attribution for regulatory reporting.

Olist — SQL Experimentation & Hypothesis Testing | *Python, DuckDB, statsmodels, SciPy*

- Ran a two-proportion z-test on 96,096 real e-commerce customers with an a-priori power analysis (alpha = 0.05, power = 0.80) sized to detect a 0.34pp difference, returning a confident null (delta = -0.03pp, p = 0.76) rather than an underpowered one.
- Wrote six window-function SQL queries (ROW_NUMBER, LAG, FIRST_VALUE) over a cohort-retention spine and found only 3.1% of customers ever reorder, redirecting retention strategy toward the first-to-second purchase.

TECHNICAL SKILLS

Languages: Python, SQL, R, TypeScript, Java
ML & Statistics: Machine learning, feature engineering, Scikit-learn, PyTorch, TensorFlow/Keras, SHAP, SMOTE, GridSearchCV, cross-validation, A/B testing, hypothesis testing, power analysis, statsmodels
NLP: spaCy, Hugging Face Transformers, vLLM, regex pipelines, RAG
Data Engineering: PySpark, dbt, Airflow, Hadoop/MapReduce/HDFS, DuckDB, Pandas, NumPy, ETL pipelines
Cloud & DevOps: AWS (EC2, S3), Google Cloud, Docker, GitHub Actions, FastAPI, REST APIs, CI/CD
Databases: PostgreSQL, pgvector, MySQL, SQLite
Visualization: Tableau, Power BI, Matplotlib, Seaborn, Streamlit